PNEUMONIA

Recovery

IN THIS ARTICLE

If you are diagnosed with pneumonia, it is important to follow your treatment plan, take steps to help your body recover, monitor your condition, and try to prevent your infection from spreading to others.

It may take time to recover from pneumonia. Some people feel better and are able to return to their normal routines in 1 to 2 weeks. For others, it can take a month or longer. Most people continue to feel tired for about a month. Talk with your healthcare provider about when you can return to your normal activities.

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Learn how to manage your pneumonia recovery. *Medical Animation Copyright © Nucleus Medical Media, All rights reserved.*

Follow your treatment plan

It is important that you take all your medicines as prescribed. If you are using antibiotics, continue to take the medicine until it is all gone. You may start to feel better before you finish the medicine, but you should continue to take it. If you stop too soon, the bacterial infection and your pneumonia may come back. The pneumonia may also become resistant to the antibiotic, making treatment more difficult.

Take steps to help your body recover

The following steps can help your body recover from pneumonia.

- **Choose heart-healthy foods**, because good nutrition helps your body recover.
- **Drink plenty of fluids** to help you stay hydrated.
- **Don't drink alcohol or use illegal drugs.** Alcohol and illegal drugs weaken your immune system and can raise the risk of pneumonia complications.
- **Don't smoke** and avoid secondhand smoke. Breathing in smoke can worsen your pneumonia. Visit Smoking and Your Heart and Your Guide to a Healthy Heart. For free help quitting smoking, you may call the National Cancer Institute's Smoking Quitline at 1-877-44U-QUIT (1-877-448-7848).
- **Get plenty of sleep.** Good quality sleep can help your body rest and improve the response of your immune system. Visit How Sleep Works to get more information.
- **Get light physical activity.** Moving around can help you regain your strength and improve your recovery. However, you may still feel short of breath. Activity that is too strenuous may make you dizzy. Talk to your provider about how much activity is right for you.
- **Sit upright** to help you feel more comfortable and breathe more easily.
- **Take a couple of deep breaths** several times a day.

Monitor your condition

Ask your provider when you should schedule follow-up care. If your symptoms have not improved, your provider may use a chest X-ray to check for other conditions that may be causing them.

Your provider may suggest pulmonary rehabilitation to help you breathe better as your lungs recover. You may also need physical therapy to help you regain your strength. Physical activity can help improve your recovery.

> Pneumonia can have long-term effects such as depression, and worsening heart and

blood vessel diseases. Call your provider if you develop these conditions, if your symptoms suddenly get worse, or if you have trouble breathing or talking.

How can pneumonia affect your health?

Often, people who have pneumonia can be successfully treated, though sometimes complications still happen. Complications are more likely if pneumonia is untreated. These complications are more common in children, older adults, and people with other serious conditions.

Learn about the complications of pneumonia. *Medical Animation Copyright © Nucleus Medical Media, All rights reserved.*

Complications of pneumonia that may be life-threatening.

- **Acute respiratory distress (ARDS) and respiratory failure** are common complications of serious pneumonia.
- **Kidney, liver, and heart damage** develops when these organs don't get enough oxygen to work properly or when your immune system responds negatively to the infection.
- **Necrotizing pneumonia** develops when your infection causes your lung tissue to die and form lung abscesses (pockets of tissue filled with pus). It also makes your pneumonia harder to treat. You may need surgery or drainage with a needle to remove the pus.
- **Pleural, or lung, disorders** occur when the tissues that cover the outside of your lungs become inflamed and the chest cavity around your lungs fills with fluid and pus.
- Sepsis happens when bacteria from your lungs gets into your blood and causes inflammation throughout your body.

Take steps to protect yourself and others

The following steps can help you prevent spreading the infection to others around you.

- **Cover your nose and mouth** while coughing or sneezing.

- **Get rid of used tissues** right away.
- **Limit contact** with family and friends.
- **Wash your hands often,** especially after coughing and sneezing.

Some people get pneumonia again and again. Tell your provider if this happens.

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Treatment

Treatment for pneumonia depends on your risk factors and how serious your pneumonia is. Many people who have pneumonia are prescribed medicine and recover at home. You may need to be treated in the hospital or an intensive care unit (ICU) if your pneumonia is serious.

Medicines

Your healthcare provider may prescribe some of the following medicines to treat your pneumonia at home or at the hospital, depending on how sick you are.

Management at home

If your pneumonia is mild, your provider may prescribe medicines or suggest over-the-counter medicines to treat it at home.

- **Antibiotics may be prescribed for bacterial pneumonia.** Most people begin to feel better after one to three days of antibiotic treatment. However, you should take antibiotics as your doctor prescribes. If you stop too soon, your pneumonia may come back.
- **Antiviral medicine is sometimes prescribed for viral pneumonia.** However, these medicines do not work against every virus that causes pneumonia.
- **Antifungal medicines are prescribed for fungal pneumonia.**
- **Over-the-counter medicines may be recommended** to treat your fever and muscle pain or help you breathe easier. Talk to your provider before taking cough or cold medicine.

Management at the hospital

If your pneumonia is serious, you may be treated in a hospital so you can get antibiotics and fluids through an intravenous (IV) line inserted into your vein. You may also get oxygen therapy to increase the amount of oxygen in your blood. If your pneumonia is very serious, you may need to be put on a ventilator.

Procedures

You may need a procedure or surgery to remove seriously infected or damaged parts of your lung. This may help you recover and may prevent your pneumonia from coming back.

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Prevention

Pneumonia can be very serious and even life-threatening. You can take a few steps to try and prevent it.

Vaccines can help prevent some types of pneumonia. Good hygiene (washing your hands often), quitting smoking, and keeping your immune system strong by getting regular physical activity and eating healthy are other ways to lower your risk of getting pneumonia.

Vaccines

Vaccines can help prevent pneumonia caused by pneumococcus bacteria or the flu virus. Vaccines cannot prevent all cases of pneumonia. However, compared to people who don't get vaccinated, those who are vaccinated and still get pneumonia tend to have:

- Fewer serious complications
- Milder infections
- Pneumonia that doesn't last as long

Pneumococcus vaccines

Two vaccines are available to prevent infections from the pneumococcus bacteria, the most common type of bacteria that causes pneumonia. Pneumococcus vaccines are especially important for people at high risk of pneumonia, including:

- Adults age 65 or older
- Children age 2 or younger
- People who have chronic (ongoing) diseases, serious long-term health problems, or weak immune systems. This may include people who have cancer, HIV, asthma, sickle cell disease, or damaged or removed spleens.
- People who smoke

For more information, visit the CDC's Pneumococcal Vaccination and Pneumococcal Vaccination: Summary of Who and When to Vaccinate.

Flu (influenza) vaccine

Your yearly flu vaccine can help prevent pneumonia caused by the flu. The flu vaccine is usually given in September through October, before flu season starts.

For more information about the flu vaccine, visit the CDC's Seasonal Influenza (Flu) Vaccination and Who Needs a Flu Vaccine and When.

Hib vaccine

Haemophilus influenzae type b (Hib) is a type of bacteria that can cause pneumonia and meningitis . The Hib vaccine is recommended for all children under 5 years old in the United States. The vaccine often is given to infants starting when they are 2 months old.

For more information about the Hib vaccine, go to the CDC's Hib Vaccination webpage.

Other ways to prevent pneumonia

You can take the following steps to help prevent pneumonia:

- **Wash your hands** with soap and water or alcohol-based hand sanitizers to kill germs.
- **Don't smoke.** Smoking prevents your lungs from properly filtering out and defending your body against germs. For information about how to quit smoking, visit Smoking and Your Heart and Your Guide to a Healthy Heart. These resources include basic information about how to quit smoking. For free help and support, you may call the National Cancer Institute's Smoking Quitline at 1-877-44U-QUIT (1-877-448-7848).
- **Keep your immune system strong.** Get plenty of physical activity and follow a healthy eating plan. Read more about heart-healthy living.
- **If you have problems swallowing,** eat smaller meals of thickened food and sleep with the head of your bed raised up. These steps can help you avoid getting food, drink, or saliva into your lungs.
- **If you have a planned surgery,** your provider may recommend that you don't eat for 8 hours or drink liquids for 2 hours before your surgery. This can help prevent food or drink from getting into your airway while you are sedated.
- **If your immune system is impaired or weakened,** your provider may recommend you take antibiotics to prevent bacteria from growing in your lungs.

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Causes and Risk Factors

Most of the time your body filters germs out of the air that you breathe. Sometimes germs, such as bacteria, viruses, or fungi, get into your lungs and cause infections.

When these germs get into your lungs, your immune system, which is your body's natural defense against germs, goes into action. Immune cells attack the germs and may cause Inflammation of your air sacs, or alveoli. Inflammation can cause your air sacs to fill up with fluid and pus and cause pneumonia symptoms. Learn about how the lungs work.

Learn About the Causes of Pneumonia

NHLBI

01:36

Watch this video to learn more about the causes of pneumonia. *Medical Animation Copyright © Nucleus Medical Media, All rights reserved.*

Causes of pneumonia

Bacteria

Bacteria are a common cause of pneumonia in adults. Many types of bacteria can cause pneumonia, but *Streptococcus pneumoniae* (also called pneumococcus bacteria) is the most common cause in the United States.

Some bacteria cause pneumonia with different symptoms or other characteristics than the usual pneumonia. This infection is called atypical pneumonia. For example, *Mycoplasma* *pneumoniae* causes a mild form of pneumonia often called "walking pneumonia." *Legionella pneumophila* causes a severe type of pneumonia called Legionnaires' disease. Bacterial pneumonia can develop on its own or after you have a cold or the flu.

Viruses

Viruses that infect your lungs and airways can cause pneumonia. The flu (influenza virus) and the common cold (rhinovirus) are the most common causes of viral pneumonia in adults. Respiratory syncytial virus (RSV) is the most common cause of viral pneumonia in young children.

Many other viruses can cause pneumonia, including SARS-CoV-2, the virus that causes COVID-19. Our video about how SARS-CoV-2 affects the lungs talks about this.

Fungi

Fungi such as *Pneumocystis jirovecii* may cause pneumonia, especially for people who have weakened immune systems. Some fungi found in the soil in the southwestern United States and in the Ohio and Mississippi River valleys can also cause pneumonia.

Risk factors

Your risk for pneumonia may be higher because of your age, environment, lifestyle habits, and other medical conditions.

Age

Pneumonia can affect people of all ages. However, two age groups are at higher risk of developing pneumonia and having more serious pneumonia.

- **Babies and children, 2 years old or younger,** are at higher risk because their immune systems are still developing. This risk is higher for premature babies.
- **Older adults, age 65 or older,** are also at higher risk because their immune systems generally weaken as people age. Older adults are also more likely to have other chronic (long-term) health conditions that raise the risk of pneumonia.

Babies, children, and older adults who do not get the recommended vaccines to prevent pneumonia have an even higher risk.

Environment or occupation

Most people get pneumonia when they catch an infection from someone else in their community. Your chance of getting pneumonia is higher if you live or spend a lot of time in a crowded place such as military barracks, prison, homeless shelters, or nursing homes.

Your risk is also higher if you regularly breathe in air pollution or toxic fumes.

Some germs that cause pneumonia can infect birds and other animals. You are most likely to encounter these germs if you work in a chicken or turkey processing center, pet shop, or veterinary clinic.

Lifestyle habits

- **Smoking cigarettes** can make you less able to clear mucus from your airways.
- **Using drugs or alcohol** can weaken your immune system. You are also more likely to accidentally breathe in saliva or vomit into your windpipe if you are sedated or unconscious from an overdose.

Other medical conditions

You may have an increased risk of pneumonia if you have any of the following medical conditions.

- **Brain disorders,** such as a stroke, a head injury, dementia, or Parkinson's disease can affect your ability to cough or swallow. This can lead to food, drink, vomit, or saliva going down your windpipe instead of your esophagus and getting into your lungs.
- **Conditions that weaken your immune system** may also increase your risk. These include pregnancy, HIV/AIDS, or an organ or bone marrow transplant. Chemotherapy, which is used to treat cancer, and long-term use of steroid medicines can also weaken your immune system.
- **Critical diseases that require hospitalization,** including receiving treatment in a hospital intensive care unit, can raise your risk of hospital-acquired pneumonia. Your risk is higher if you cannot move around much or are sedated or unconscious. Using a ventilator raises the risk of a type called ventilator-associated pneumonia.
- **Lung diseases,** such as asthma, bronchiectasis, cystic fibrosis, or COPD, also increase your pneumonia risk.
- **Other serious conditions,** such as malnutrition, diabetes, heart failure, sickle cell disease, or liver or kidney disease, are additional risk factors.

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Diagnosis

Your healthcare provider will diagnose pneumonia based on your medical history, a physical exam, and test results. Sometimes pneumonia is hard to diagnose because your symptoms may be the same as a cold or flu. You may not realize that your condition is more serious until it lasts longer than these other conditions.

Medical history and physical exam

Your provider will ask about your symptoms and when they began. They will also ask whether you have any risk factors for pneumonia. You may also be asked about:

- Exposure to sick people at home, school, or work or in a hospital
- Flu or pneumonia vaccinations
- Medicines you take
- Past and current medical conditions and whether any have gotten worse recently
- Recent travel
- Exposure to birds and other animals
- Smoking

During your physical exam, your provider will check your temperature and listen to your lungs with a stethoscope.

Diagnostic tests and procedures

If your provider thinks you have pneumonia, he or she may do one or more of the following tests.

- **A chest X-ray** looks for inflammation in your lungs. A chest X-ray is often used to diagnose pneumonia.
- **Blood tests**, such as a complete blood count (CBC) see whether your immune system is fighting an infection.

- **Pulse oximetry** measures how much oxygen is in your blood. Pneumonia can keep your lungs from getting enough oxygen into your blood. To measure the levels, a small sensor called a pulse oximeter is attached to your finger or ear.

If you are in the hospital, have serious symptoms, are older, or have other health problems, your provider may do other tests to diagnose pneumonia.

- **A blood gas test may be done** if you are very sick. For this test, your provider measures your blood oxygen levels using a blood sample from an artery, usually in your wrist. This is called an arterial blood gas test.
- **A sputum test, using a sample of sputum (spit) or mucus from your cough, may be used to** find out what germ is causing your pneumonia.
- **A blood culture** test can identify the germ causing your pneumonia and also show whether a bacterial infection has spread to your blood.
- **A polymerase chain reaction (PCR) test** quickly checks your blood or sputum sample to find the DNA of germs that cause pneumonia.
- **A bronchoscopy** looks inside your airways. If your treatment is not working well, this procedure may be needed. At the same time, your doctor may also collect samples of your lung tissue and fluid from your lungs to help find the cause of your pneumonia.
- **A chest computed tomography (CT) scan** can show how much of your lungs are affected by pneumonia. It can also show whether you have complications such as lung abscesses or pleural disorders. A CT scan shows more detail than a chest X-ray.
- **A pleural fluid culture can be taken using a procedure called** thoracentesis, which is when a doctor uses a needle to take a sample of fluid from the pleural space between your lungs and chest wall. The fluid is then tested for bacteria.

Participate in a study

We lead or sponsor many trials and studies on pneumonia. See if you or someone you love is eligible to join.

Find pneumonia studies

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Symptoms

The symptoms of pneumonia can be mild or serious. Young children, older adults, and people who have serious health conditions are at risk for developing more serious pneumonia or life-threatening complications.

The symptoms of pneumonia may include:

- Chest pain when you breathe or cough
- Chills
- Cough with or without mucus
- Fever
- Low oxygen levels in your blood, measured with a pulse oximeter
- Shortness of breath

You may also have other symptoms, including a headache, muscle pain, extreme tiredness, nausea (feeling sick to your stomach), vomiting, and diarrhea.

Older adults and people who have serious illnesses or weakened immune systems may not have the typical symptoms. They may have a lower-than-normal temperature instead of a fever. Older adults who have pneumonia may feel weak or suddenly confused.

Sometimes babies don’t have typical symptoms either. They may vomit, have a fever, cough, or appear restless or tired and without energy. Babies may also show the following signs of breathing problems:

- Bluish tone to the skin and lips
- Grunting
- Pulling inward of the muscles between the ribs when breathing
- Rapid breathing
- Widening of the nostrils with each breath

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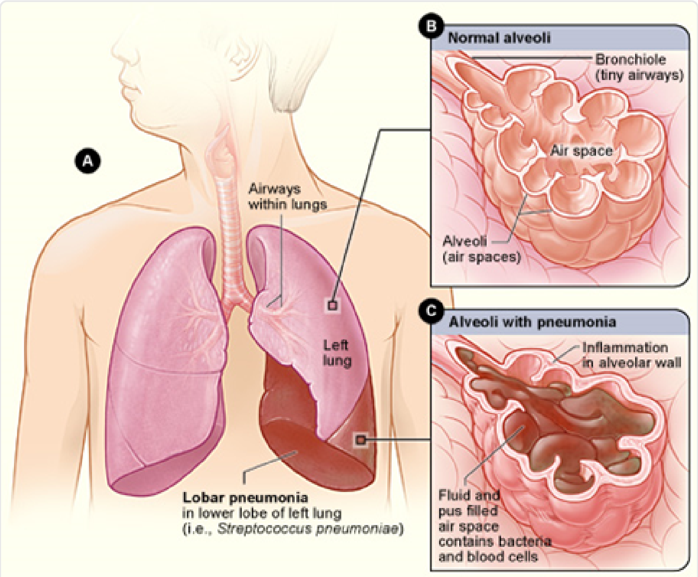
What Is Pneumonia?

Pneumonia is an infection that affects one or both lungs. It causes the air sacs, or alveoli, of the lungs to fill up with fluid or pus. Bacteria, viruses, or fungi may cause pneumonia. Symptoms can range from mild to serious and may include a cough with or without mucus (a slimy substance), fever, chills, and trouble breathing. How serious your pneumonia is depends on your age, your overall health, and what caused your infection.

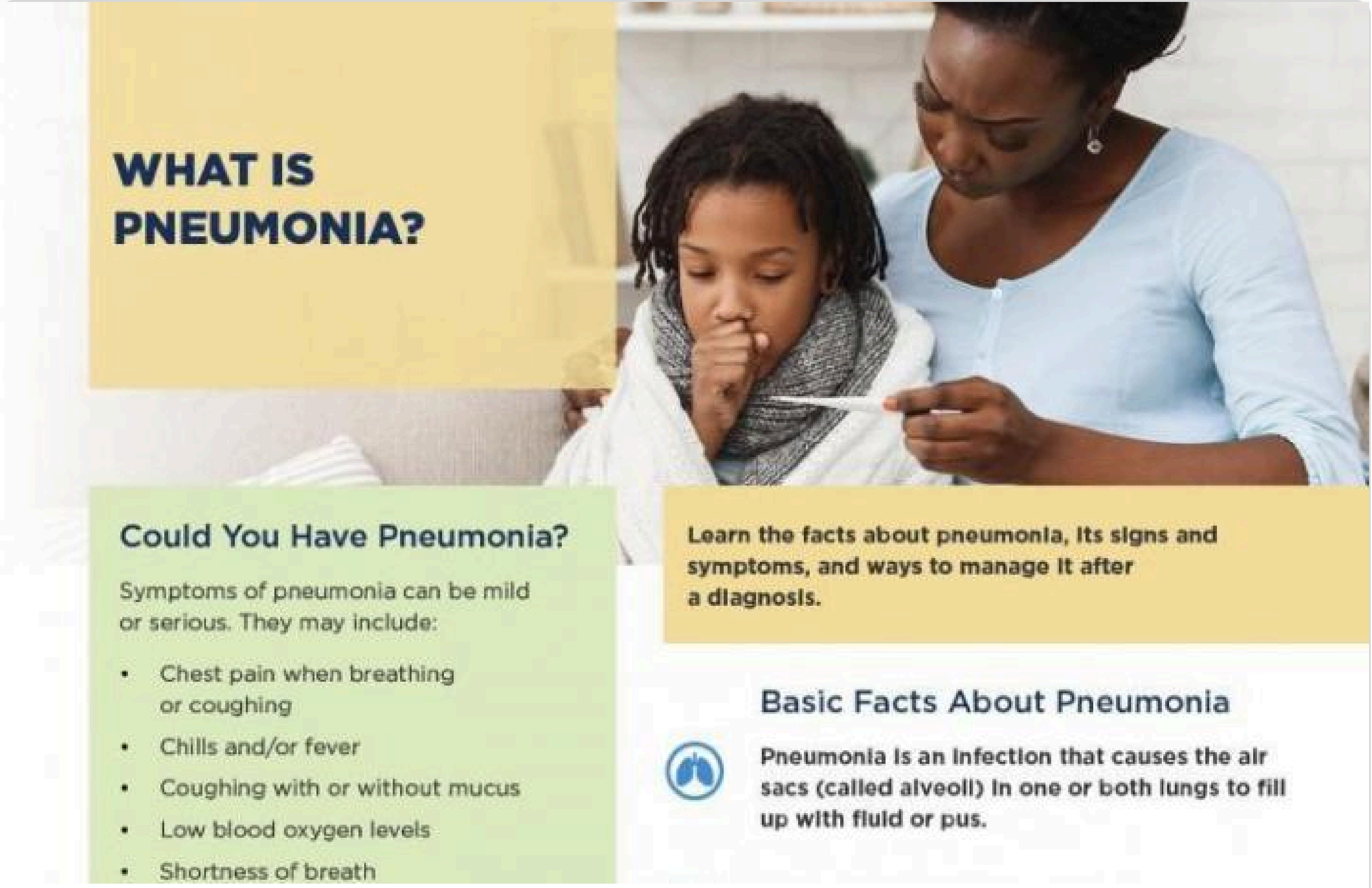


Pneumonia, caused by bacteria. Figure A shows pneumonia affecting part of the left lung. Figure B shows healthy alveoli (air sacs). Figure C shows alveoli filled with mucus.

To diagnose pneumonia, your healthcare provider will review your medical history, perform a physical exam, and order diagnostic tests such as a chest X-ray. This information can help determine what type of pneumonia you have.

Treatment for pneumonia may include antibiotic, viral, or fungal medicines. It may take several weeks to recover from pneumonia. If your symptoms get worse, you should see a healthcare provider right away. If you have severe pneumonia, you may need to go to the hospital for antibiotics given through an intravenous (IV) line and oxygen therapy.

Some types of pneumonia can be prevented by vaccines. Good hygiene and heart-healthy living can also lower your risk for pneumonia.



FACT SHEET

What Is Pneumonia?

This fact sheet describes pneumonia, including how it affects breathing, what causes it, and how it's diagnosed and treated.

[View the fact sheet](#)

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